

Session 01

Q: 01What is Programming? Topics: Define “program” and “programming language.” Explain compilers and interpreters. Discuss importance of algorithms and flowcharts. **Trainer Activity:** Draw a flowchart for “Ordering food online.”

=> Programming:-

Programming is the Process of creating step by step instructions that tell a computer how to perform a task or solve a problem.

Program:-

A program is set of structured instructions or a planned sequence of activities designed to achieve a specific goal.

Programming Language:-

A programming language is a formal set of rules, syntax and symbols used to write instructions that a computer can understand and execute.

A **Compiler** and an **interpreter** are tools that translate human readable programming code like C++, Python into machine readable language binary machine language.

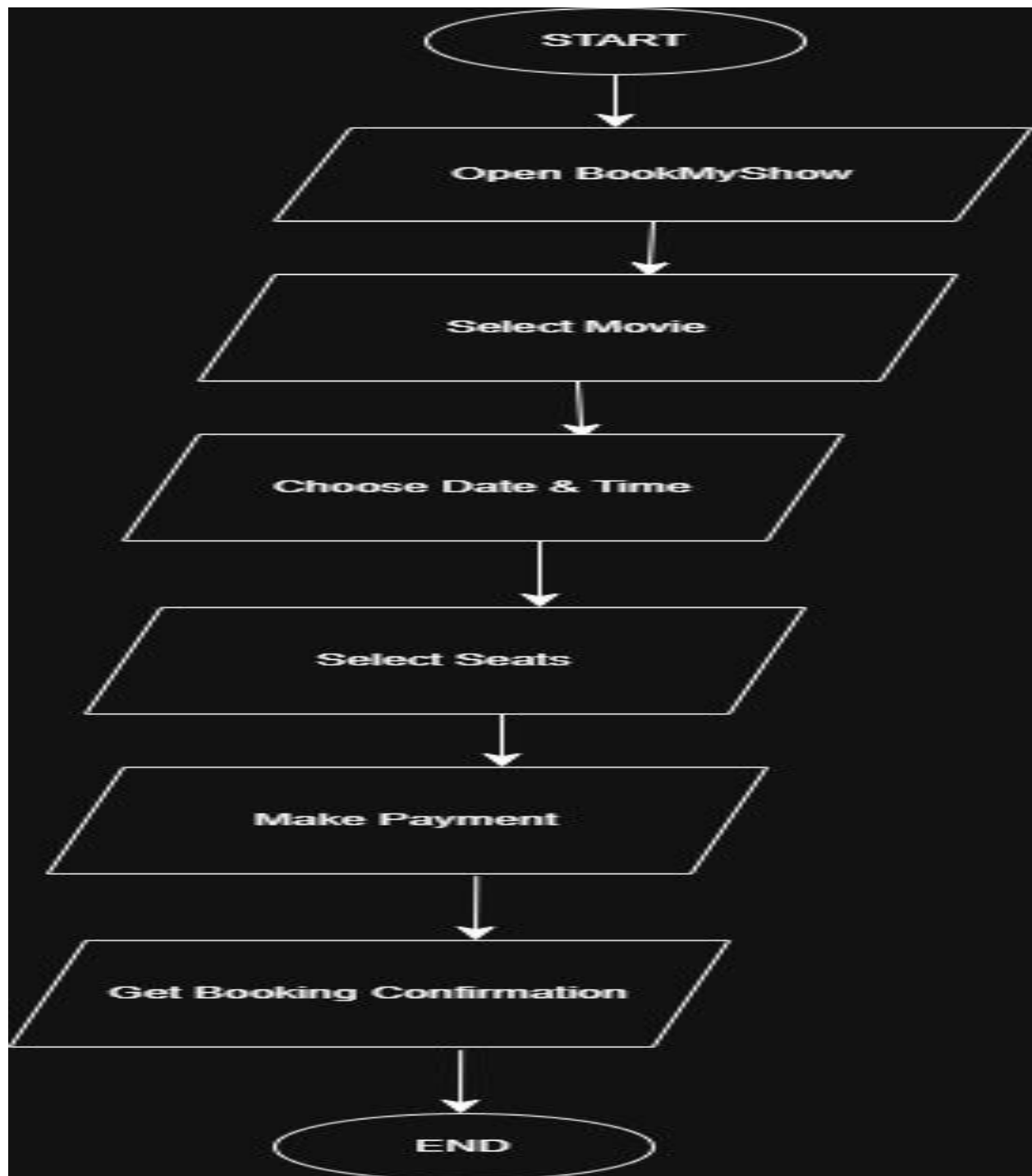
Compiler:-

Translates the entire source code all at once into an executable file *before* running it. It catches all errors beforehand but runs very fast upon execution . Languages like C, C++, and Rust use compilers

Interpreter:-

Translates and executes the code line-by-line in real-time *while* the program is running. It stops executing whenever it hits an error, making debugging easier. Languages like Python and JavaScript use interpreters.

Flowchart for “Movie ticket online:-



Session 6: Loops (For, While, Do-While) Topics: Difference between entry-controlled and exit-controlled loops. Use cases (e.g., menu-driven programs).Example: Countdown timer using loops.

Entry Controlled:-

First check the condition after than execute code.

Ex:-

While and for loop.

Exit Controlled:-

Execute code first then check the condition

Ex:-

Do_while.